

1.6 Bonding

Students will be assessed on their ability to:

- 1 Ionic bonding
 - a recall and interpret evidence for the existence of ions, limited to physical properties of ionic compounds, electron density maps and the migration of ions, eg electrolysis of aqueous copper chromate(VI)
 - b describe the formation of ions in terms of electron loss or gain
 - c draw electron configuration diagrams of cations and anions using dots or crosses to represent electrons
 - d describe ionic crystals as giant lattices of ions
 - e describe ionic bonding as the result of strong net electrostatic attraction between ions
 - f recall trends in ionic radii down a group and for a set of isoelectronic ions, eg N^{3-} to Al^{3+}
 - g recall the stages involved in the formation of a solid ionic crystal from its elements and that this leads to a measure value for the lattice energy (students will not be expected to draw the full Born-Haber cycles)

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- h test the ionic model for ionic bonding of a particular compound by comparison of lattice energies obtained from the experimental values used in Born-Haber cycles, with provided values calculated from electrostatic theory
 - i explain the meaning of the term polarization as applied to ions
 - j demonstrate an understanding that the polarizing power of a cation depends on its radius and charge, and the polarizability of an anion depends on its size
 - k demonstrate an understanding that polarization of anions by cations leads to some covalency in an ionic bond, based on evidence from the Born-Haber cycle
 - l use values calculated for standard heats of formation based on Born-Haber cycles to explain why particular ionic compounds exist, eg the relative stability of MgCl_2 over MgCl or MgCl_3 and NaCl over NaCl_2 .

2 Covalent bonding

- a demonstrate an understanding that covalent bonding is strong and arises from the electrostatic attraction between the nucleus and the electrons which are between the nuclei, based on the evidence:
 - i the physical properties of giant atomic structures
 - ii electron density maps for simple molecules
- b draw electron configuration diagrams for simple covalently bonded molecules, including those with multiple bonds and dative covalent bonds, using dots or crosses to represent electrons.

3 Metallic bonding

- a demonstrate an understanding that metals consist of giant lattices of metal ions in a sea of delocalised electrons
- b describe metallic bonding as the strong attraction between metal ions and the sea of delocalised electrons
- c use the models in 1.6.3a and 1.6.3b to interpret simple properties of metals, eg conductivity and melting temperatures.